

Cryptography Questions and Answers

1. In RSA, the private exponent d is chosen so that it satisfies:

- A. d is simply equal to $\phi(n)$ divided by e , with any remainder discarded
- B. d must equal n minus e , computed without any modular reduction
- C. d is the multiplicative inverse of e , modulo $\phi(n)$**
- D. d is chosen as any prime number larger than both p and q

Answer: C. d is the multiplicative inverse of e , modulo $\phi(n)$

2. NIST's selection of Rijndael as AES was notable because it was:

- A. Chosen without any competing candidate algorithms being considered
- B. The result of an open, multi-round public competition rather than an algorithm designed internally by the government**
- C. The first cipher ever to be patented by a government standards body
- D. Selected purely on the basis of being the fastest candidate in hardware

Answer: B. The result of an open, multi-round public competition rather than an algorithm designed internally by the government

3. Output Feedback (OFB) mode differs from CFB mode chiefly in that OFB:

- A. Feeds back the preceding encryption output itself, rather than the ciphertext, into the next encryption step**
- B. Does not require an initialization vector at all
- C. Cannot be used to convert a block cipher into a stream-like cipher
- D. Always operates on smaller sub-block units than CFB does

Answer: A. Feeds back the preceding encryption output itself, rather than the ciphertext, into the next encryption step

4. Why is factoring huge numbers considered computationally hard even when a shortcut like the shared-prime-factor rule can quickly find a gcd once factorizations are already known?

- A. Because the gcd shortcut only ever works for numbers smaller than 1,000
- B. Because gcd computation requires knowledge of the discrete logarithm first
- C. That shortcut assumes the factorizations are already available; obtaining those factorizations for very large numbers in the first place is itself the hard step**
- D. Because factoring and computing a gcd are mathematically the identical operation

Answer: C. That shortcut assumes the factorizations are already available; obtaining those factorizations for very large numbers in the first place is itself the hard step

5. The AES State array arranges the 16 bytes of a block into a matrix of:

- A. 8 rows by 2 columns
- B. 2 rows by 8 columns
- C. 4 rows by 4 columns**
- D. 16 rows by 1 column

Answer: C. 4 rows by 4 columns

6. Using $p = 17$, $q = 11$, $e = 7$, and $d = 23$ (so $n = 187$ and $\phi(n) = 160$), encrypting $M = 88$ gives a ciphertext C computed as:

- A. $C = 88^7 \bmod 187 = 11$**
- B. $C = 11^7 \bmod 187 = 88$
- C. $C = 88^{23} \bmod 187 = 11$
- D. $C = 88^7 \bmod 160 = 11$

Answer: A. $C = 88^7 \bmod 187 = 11$

7. The classic, simple algorithm defined by the recurrence $X_{n+1} = (aX_n + c) \bmod m$ is called a ____ generator.

Answer: Linear-Congruential

8. Decryption of a MixColumns-transformed column uses:

- A. InvMixColumns, multiplying by the inverse of the fixed matrix used during encryption**
- B. No matrix at all, since MixColumns is skipped automatically during decryption
- C. A completely different, randomly generated matrix chosen for each session
- D. The identical MixColumns matrix used during encryption, applied a second time

Answer: A. InvMixColumns, multiplying by the inverse of the fixed matrix used during encryption

9. For a positive integer n , arithmetic performed modulo n confines every result to the range:

- A. $\{0, 1, \dots, 2n-1\}$, doubling the usual modular range for security
- B. $\{1, 2, \dots, n\}$, excluding zero entirely from every calculation
- C. $\{-n, \dots, 0, \dots, n\}$, allowing both positive and negative remainders freely
- D. $\{0, 1, \dots, n-1\}$, wrapping around whenever a value would leave that range**

Answer: D. $\{0, 1, \dots, n-1\}$, wrapping around whenever a value would leave that range

10. Given $x \equiv 2 \pmod{3}$ and $x \equiv 3 \pmod{5}$, the Chinese Remainder Theorem gives the unique solution in the range 0 to 14 as:

A. $x = 11$

B. $x = 8$

C. $x = 13$

D. $x = 2$

Answer: B. $x = 8$

11. A number-theoretic PRNG whose security rests on the difficulty of factoring $n = p \times q$, the same hard problem underlying RSA, is called the ____ generator.

Answer: Blum-Blum-Shub

12. The recommended padding scheme that randomizes RSA plaintext before encryption, defending against chosen-ciphertext attacks, is called ____.

Answer: OAEP

13. After the KSA, RC4's Pseudo-Random Generation Algorithm (PRGA) produces the keystream by:

A. Continuously swapping entries within the permuted state array S and combining selected entries to output one byte at a time

B. Applying the SHA-1 hash function repeatedly to the previous keystream byte

C. Reading the secret key directly, byte by byte, without any further processing

D. Encrypting a running counter with a block cipher and outputting the ciphertext bytes

Answer: A. Continuously swapping entries within the permuted state array S and combining selected entries to output one byte at a time

14. Fermat's Little Theorem states that, for a prime p and an integer a not divisible by p :

A. a^p is congruent to a modulo $(p-1)$

B. $a^{(p-1)}$ is congruent to 1 modulo p

C. $a^{(p+1)}$ is congruent to a modulo p

D. $p^{(a-1)}$ is congruent to 1 modulo a

Answer: B. $a^{(p-1)}$ is congruent to 1 modulo p

15. In RSA, encryption of message M is computed as $C = M^e \pmod{n}$, and decryption of ciphertext C is computed as $M = C^{\text{____}} \pmod{n}$.

Answer: d

16. The Intel Digital Random Number Generator (DRNG), used in Intel processors since 2012, combines which stages in its pipeline?

A. A thermal-noise entropy source, a bias-removing conditioner, and a CTR_DRBG stage exposed through the RDRAND instruction

B. A user-supplied password hashed once with SHA-1 and output directly

C. Two independent TRNGs whose raw, unconditioned outputs are simply concatenated

D. An LCG seeded from the system clock, followed directly by RC4 encryption

Answer: A. A thermal-noise entropy source, a bias-removing conditioner, and a CTR_DRBG stage exposed through the RDRAND instruction

17. Counter (CTR) mode generates its keystream by:

A. Chaining each block's ciphertext into the encryption of the following block

B. Encrypting a counter value that increments for each block, then XORing the result with the plaintext

C. Reusing the same encrypted value as the keystream for every block in a message

D. Encrypting the plaintext directly with no counter or feedback of any kind

Answer: B. Encrypting a counter value that increments for each block, then XORing the result with the plaintext

18. A base g whose successive powers modulo a prime p cycle through every nonzero remainder before repeating is called a ____ of p .

Answer: primitive root

19. The square-and-multiply technique for RSA is valuable because it computes a large modular exponentiation such as x^{16} using:

A. Addition alone, avoiding multiplication entirely

B. Repeated squaring, needing only a handful of multiplications instead of performing the full exponent's worth

C. A random guess-and-check process repeated until the result matches

D. A single lookup table indexed by the exponent value

Answer: B. Repeated squaring, needing only a handful of multiplications instead of performing the full exponent's worth

20. In Cipher Block Chaining (CBC) mode, the input to the encryption function for a given block is:

A. The XOR of the current plaintext block with the next plaintext block

B. The previous plaintext block encrypted a second time

C. The XOR of the current plaintext block with the previous ciphertext block (or the IV, for the first block)

D. The plaintext block alone, exactly as in ECB mode

Answer: C. The XOR of the current plaintext block with the previous ciphertext block (or the IV, for the first block)

21. The Miller-Rabin primality test is best described as:

A. A method for factoring a composite number into its two prime factors

B. A fast probabilistic test that either proves a candidate is composite or leaves it as only probably prime

C. A technique for generating the private key directly from a public RSA modulus

D. A deterministic test, like trial division, that always proves primality with total certainty

Answer: B. A fast probabilistic test that either proves a candidate is composite or leaves it as only probably prime

22. A mode of operation that allows random access to any individual block without first decrypting all preceding blocks is best exemplified by:

A. ECB mode used together with an initialization vector

B. CTR mode, since each block's keystream depends only on its own counter value

C. CFB mode, since it processes data in variable-size segments

D. CBC mode, since each block depends on the ciphertext immediately before it

Answer: B. CTR mode, since each block's keystream depends only on its own counter value

23. In the prime field $GF(p)$, where p is prime, every nonzero element has a multiplicative inverse precisely because:

A. p is always chosen to be an even number greater than 2

B. Multiplication in $GF(p)$ is defined without ever using modular reduction

C. $GF(p)$ is required to contain exactly two elements by definition

D. p is prime, so no nonzero element shares a common factor with p

Answer: D. p is prime, so no nonzero element shares a common factor with p

24. AES is generally considered secure today mainly because:

A. Its block size grows automatically as computing power increases

B. Its algorithm and S-box design have never been published or reviewed publicly

C. It uses a Feistel structure, which is inherently immune to differential cryptanalysis

D. Its 2^{128} (or larger) keyspace makes brute force infeasible, and decades of public cryptanalysis have found no effective shortcut attack

Answer: D. Its 2^{128} (or larger) keyspace makes brute force infeasible, and decades of public cryptanalysis have found no effective shortcut attack

25. Using the Chinese Remainder Theorem to split RSA decryption into two smaller modular exponentiations, one modulo p and one modulo q , speeds up decryption by roughly a factor of ____.

Answer: 4

26. In $GF(2^m)$, addition of two field elements is performed using the bitwise operation ____.

Answer: XOR

27. Which of the following was NOT one of the finalist algorithms in the AES selection process?

- A. Twofish
- B. MARS
- C. Blowfish**
- D. Serpent

Answer: C. Blowfish

28. DES processes plaintext through which overall sequence of steps?

- A. A single large substitution table applied directly to the full 64-bit block
- B. An initial permutation, sixteen Feistel rounds, a swap of the two halves, then the inverse initial permutation**
- C. Byte substitution, row shifting, column mixing, and a round-key XOR, repeated ten times
- D. A key expansion, eight parallel Feistel rounds, then a single final substitution

Answer: B. An initial permutation, sixteen Feistel rounds, a swap of the two halves, then the inverse initial permutation

29. The 4×4 byte matrix that AES loads the plaintext block into, and which every round transformation operates on, is called the ____.

Answer: state

30. The security of Diffie-Hellman, ElGamal encryption, and the Digital Signature Algorithm all ultimately rest on the assumed hardness of:

- A. Finding a collision in a cryptographic hash function
- B. Factoring a large composite number into its two prime factors
- C. The discrete logarithm problem**
- D. Solving simultaneous linear equations over the integers

Answer: C. The discrete logarithm problem

31. $GF(2^8)$ arithmetic is directly used inside which widely deployed algorithm?

- A. The Euclidean Algorithm, when computing a greatest common divisor
- B. RSA, when selecting the two large prime factors of its modulus
- C. AES, in its byte substitution and column mixing steps**
- D. The plain Caesar cipher, in choosing its shift amount

Answer: C. AES, in its byte substitution and column mixing steps

32. An algebraic structure with two operations in which every nonzero element has a multiplicative inverse (such as $GF(p)$ for a prime p) is called a ____.

Answer: field

33. Shannon's principle of confusion is achieved mainly by:

- A. Spreading each plaintext bit's influence across the whole ciphertext block
- B. Choosing a block size larger than the key size
- C. Repeating the plaintext block several times before encryption begins
- D. Making the relationship between the ciphertext and the key as complex as possible, typically through substitution**

Answer: D. Making the relationship between the ciphertext and the key as complex as possible, typically through substitution

34. AES is best classified structurally as a:

- A. Substitution-permutation network operating on the full block every round, not a Feistel cipher**
- B. Hash-based construction with no dependency on a secret key
- C. Stream cipher that produces one keystream byte at a time
- D. Feistel cipher that splits its block into two halves each round

Answer: A. Substitution-permutation network operating on the full block every round, not a Feistel cipher

35. The standard practical fix for the Diffie-Hellman man-in-the-middle vulnerability is to add:

A. Digital signatures and public-key certificates to authenticate each party's exchanged value

B. A symmetric pre-shared key used instead of the discrete logarithm exchange

C. Multiple additional unauthenticated rounds of the identical exchange

D. A longer prime modulus q , with no other changes required to the protocol

Answer: A. Digital signatures and public-key certificates to authenticate each party's exchanged value

36. In Electronic Codebook (ECB) mode, each plaintext block is:

A. XORed with the previous ciphertext block before being encrypted

B. Split into two halves and processed through a Feistel-like structure

C. Encrypted independently with the same key, with no chaining or feedback between blocks

D. Combined with a running counter before being encrypted

Answer: C. Encrypted independently with the same key, with no chaining or feedback between blocks

37. The Euclidean Algorithm computes $\gcd(a, b)$ by:

A. Multiplying a and b together and then dividing by their least common multiple

B. Repeatedly dividing and keeping the remainder until a remainder of zero is reached

C. Testing every integer from 1 up to the smaller of a and b for common divisibility

D. Factoring both a and b completely and comparing their prime factorizations

Answer: B. Repeatedly dividing and keeping the remainder until a remainder of zero is reached

38. The points on an elliptic curve, together with a defined addition rule and a point at infinity, form:

A. An abelian group, satisfying closure, associativity, an identity element, inverses, and commutativity

B. A Feistel structure, alternating substitution and permutation at each point

C. A finite field in the same sense as $GF(p)$, with every point having a multiplicative inverse

D. A simple ring with no guarantee that every point has an inverse

Answer: A. An abelian group, satisfying closure, associativity, an identity element, inverses, and commutativity

39. Digital signatures are only ever applied to a hash of the message, never its content.

A. Digital signatures are only ever applied to a hash of the message, never its content

B. Anyone who has the sender's public key can decrypt and read the message

C. The message is never actually transformed, only tagged with a plaintext label

D. The private key used is always shorter than the key used for encryption alone

Answer: A. Digital signatures are only ever applied to a hash of the message, never its content

40. Timing-attack research on DES found that measuring decryption time can reveal:

A. Nothing useful at all, since DES executes in constant time by design

B. The plaintext directly, without needing to know anything about the key

C. The complete 56-bit key after only a handful of timed decryptions

D. The Hamming weight of the key (how many key bits are set to one), but not the key itself

Answer: D. The Hamming weight of the key (how many key bits are set to one), but not the key itself

41. In the terminology Stallings uses for classical and modern ciphers, permutation is best described as an operation that:

A. Compresses the plaintext into a shorter, fixed-length representation

B. Rearranges the order of plaintext elements without adding, removing, or replacing any of them

C. Splits the plaintext into independent blocks that are encrypted with different keys

D. Replaces each plaintext element with a different symbol drawn from a fixed alphabet

Answer: B. Rearranges the order of plaintext elements without adding, removing, or replacing any of them

42. Which requirement is essential when parameterizing an RSA-based PRNG such as Micali-Schnorr?

A. e must be chosen equal to the private exponent d used for RSA decryption

B. $\gcd(e, \phi(n)) = 1$, mirroring the same requirement used for ordinary RSA key generation

C. The modulus n must be chosen to be a small prime number, unlike ordinary RSA

D. The output length r must always exceed the modulus n 's bit length

Answer: B. $\gcd(e, \phi(n)) = 1$, mirroring the same requirement used for ordinary RSA key generation

43. For the prime 15's totient, $\phi(15) = \phi(3) \times \phi(5) = 2 \times 4 = 8$. This calculation relies on the fact that:

A. ϕ is multiplicative over the product of two distinct primes, since $\phi(pq) = (p-1)(q-1)$

B. $\phi(n)$ is always equal to n divided by 2, rounded down to the nearest integer

C. 15 itself is prime, so $\phi(15)$ is always equal to 14 by definition

D. Every factor of 15 must individually be tested for primality first

Answer: A. ϕ is multiplicative over the product of two distinct primes, since $\phi(pq) = (p-1)(q-1)$

44. Historically, the clearest and most practical weakness of DES has been:

A. Its 56-bit key length, which modern hardware can exhaustively search

B. A lack of an inverse initial permutation, making decryption impossible

C. A publicly proven mathematical flaw in the design of its S-boxes

D. Its inability to be combined with any block cipher mode of operation

Answer: A. Its 56-bit key length, which modern hardware can exhaustively search

45. In evaluating candidate round functions F for a block cipher, the strict avalanche criterion (SAC) specifically requires that:

A. Each output bit should change with probability one-half whenever a single input bit is inverted

B. The round function should use exactly the same number of rounds as the key length in bits

C. Every output bit should be a linear combination of the input bits

D. The ciphertext should always be exactly the same length as the plaintext

Answer: A. Each output bit should change with probability one-half whenever a single input bit is inverted

46. Because OFB feeds back the encryption output rather than the ciphertext, it offers the practical advantage that:

A. It becomes fully immune to any form of cryptanalysis

B. It automatically compresses the ciphertext to a smaller size than the plaintext

C. A bit error introduced during transmission affects only the corresponding plaintext bit, without cascading into later blocks

D. No initialization vector or key is required to begin decryption

Answer: C. A bit error introduced during transmission affects only the corresponding plaintext bit, without cascading into later blocks

47. The ElGamal cryptosystem, published in 1984, is most accurately described as:

- A. An improved factoring algorithm used to attack RSA moduli directly
- B. A hash-based signature scheme that does not rely on any modular exponentiation
- C. A public-key encryption scheme closely related to Diffie-Hellman, resting on the same discrete-logarithm hardness assumption**
- D. A symmetric block cipher unrelated to any discrete-logarithm-based scheme

Answer: C. A public-key encryption scheme closely related to Diffie-Hellman, resting on the same discrete-logarithm hardness assumption

48. RSA was developed in 1977 by Ron Rivest, Adi Shamir, and Len _____, whose initials give the algorithm its name.

Answer: Adleman

49. Asymmetric ciphers such as RSA and ECC are not typically used to generate long, open-ended pseudorandom bit streams because:

- A. They require a hardware TRNG to be present, unlike every symmetric-cipher-based PRNG
- B. Asymmetric operations are comparatively slow, so they are better suited to producing a short pseudorandom function output per invocation**
- C. They are mathematically incapable of producing any output that looks statistically random
- D. Their output is always shorter than the modulus n , regardless of the number of invocations

Answer: B. Asymmetric operations are comparatively slow, so they are better suited to producing a short pseudorandom function output per invocation

50. XTS-AES ties encryption to a particular physical storage location by using:

- A. A running counter that resets to zero once per day
- B. A separate hash function computed over the entire disk
- C. A second, independently generated AES key for every sector
- D. A tweak value derived from the sector or block address, alongside the key**

Answer: D. A tweak value derived from the sector or block address, alongside the key

51. The final round of AES encryption differs from the earlier rounds in that it:

- A. Omits the AddRoundKey transformation entirely
- B. Repeats SubBytes twice in succession
- C. Omits the MixColumns transformation**
- D. Uses a different, smaller S-box than earlier rounds

Answer: C. Omits the MixColumns transformation

52. A defining advantage of the Feistel structure is that:

- A. No subkey schedule is required, since the same key is reused every round
- B. Decryption uses the identical algorithm as encryption, only with the subkeys applied in reverse order**
- C. The round function F must itself be a reversible, invertible operation
- D. Encryption and decryption each require a completely separate algorithm

Answer: B. Decryption uses the identical algorithm as encryption, only with the subkeys applied in reverse order

53. Triple DES (3DES), typically applying Encrypt-Decrypt-Encrypt with two or three keys, achieves an effective key length of up to:

- A. 168 bits, using three independent keys**
- B. 56 bits, the same as single DES
- C. 224 bits, matching twice the length of AES-256's key
- D. 112 bits, even when three independent keys are used

Answer: A. 168 bits, using three independent keys

54. An RSA-based pseudorandom number generator that encrypts its internal state at each stage, feeding back some bits and outputting the rest, is called the ____ PRNG.

Answer: RSA

55. RC4 begins with a Key Scheduling Algorithm (KSA) that:

- A. Fixes the state array S to a constant, unchanging table for every key
- B. Directly encrypts the plaintext without ever touching a state array
- C. Uses the secret key to scramble a 256-byte state array S into a key-dependent permutation**
- D. Generates a brand-new AES key for every byte of output produced

Answer: C. Uses the secret key to scramble a 256-byte state array S into a key-dependent permutation

56. The Fundamental Theorem of Arithmetic guarantees that every integer greater than 1:

- A. Shares at least one prime factor with every other integer greater than 1
- B. Can be written as a product of primes in exactly one way, ignoring the order of the factors**

- C. Is itself either prime or a perfect square, with no other possibilities
- D. Has a unique representation as the sum, rather than the product, of primes

Answer: B. Can be written as a product of primes in exactly one way, ignoring the order of the factors

57. The discrete logarithm problem asks, given a base g , a prime p , and a value b , to find x such that:

- A. x^g is congruent to b modulo p
- B. $g + x$ is congruent to b modulo p
- C. x is congruent to $\gcd(g, b)$ modulo p

D. g^x is congruent to b modulo p

Answer: D. g^x is congruent to b modulo p

58. In Diffie-Hellman, both public parameters q (a prime) and α (a primitive root of q) are:

- A. Derived mathematically from each party's private exponent after the exchange
- B. Generated fresh for every single message exchanged between the parties
- C. Kept secret and known only to the two communicating parties

D. Known publicly to everyone, including any potential attacker

Answer: D. Known publicly to everyone, including any potential attacker

59. The security of Diffie-Hellman key exchange rests on the assumed difficulty of:

- A. Reversing the AES S-box without knowing the round key
- B. Computing discrete logarithms modulo a large prime**
- C. Factoring the product of two large primes
- D. Finding a collision in a cryptographic hash function

Answer: B. Computing discrete logarithms modulo a large prime

60. An integer in $\mathbb{Z}_8$ fails to have a multiplicative inverse modulo 8 precisely when it:

A. Shares a common factor greater than 1 with 8

- B. Is an odd number less than 8
- C. Is larger than half of 8
- D. Is itself equal to zero or to one

Answer: A. Shares a common factor greater than 1 with 8

61. In cryptography, randomness and unpredictability are best distinguished as follows:

- A. They are simply two different names for the exact same property of a number sequence
- B. Randomness applies only to hardware generators, while unpredictability applies only to software algorithms
- C. Randomness refers to statistical uniformity and independence, while unpredictability means future values cannot be guessed even after observing many past values**
- D. Unpredictability means the sequence never repeats, while randomness means it always repeats

Answer: C. Randomness refers to statistical uniformity and independence, while unpredictability means future values cannot be guessed even after observing many past values

62. The widely deployed stream cipher designed by Ron Rivest in 1987, later used in SSL/TLS and WEP, is called ____.

Answer: RC4

63. A key limitation of CBC mode compared to CTR mode is that CBC encryption:

- A. Cannot use an initialization vector, unlike every other mode
- B. Reveals identical plaintext blocks as identical ciphertext blocks, just like ECB
- C. Requires a completely different block cipher algorithm for each block
- D. Cannot be parallelized, because each block depends on the ciphertext of the block before it**

Answer: D. Cannot be parallelized, because each block depends on the ciphertext of the block before it

64. The attack that defeats Double DES by working forward from the plaintext and backward from the ciphertext to find a matching intermediate value is called the ____ attack.

Answer: meet-in-the-middle

65. An element a is called a primitive root of a prime p when:

- A. a raised to any power always produces the same remainder modulo p
- B. The successive powers of a , taken modulo p , cycle through every nonzero remainder before repeating**
- C. a is the smallest prime factor of p minus one
- D. a itself must also be prime and strictly greater than p

Answer: B. The successive powers of a , taken modulo p , cycle through every nonzero remainder before repeating

66. Shannon's principle of diffusion is achieved mainly by:

- A. Replacing the round function with a simple table lookup
- B. Making the ciphertext statistics depend on the key in a highly complex way
- C. Reducing the key length needed to resist brute-force search

D. Spreading the influence of each plaintext bit over many bits of the ciphertext

Answer: D. Spreading the influence of each plaintext bit over many bits of the ciphertext

67. The Diffie-Hellman protocol requires two public parameters known to everyone: a prime q and a ____ root of q .

Answer: primitive

68. Addition of two elements in $GF(2^m)$ is carried out as:

- A. Concatenation of the two bit strings into one longer string
- B. Ordinary decimal addition followed by reduction modulo m
- C. Bitwise XOR of the corresponding binary coefficients, with no carrying**
- D. Multiplication of the two elements' polynomial representations

Answer: C. Bitwise XOR of the corresponding binary coefficients, with no carrying

69. In ShiftRows, row r of the State is cyclically shifted left by:

- A. One byte for every row without exception
- B. r bytes, so row 0 is unshifted, row 1 shifts by 1, row 2 by 2, and row 3 by 3**
- C. r bits rather than r bytes, applied to every row equally
- D. A fixed 4 bytes regardless of which row it is

Answer: B. r bytes, so row 0 is unshifted, row 1 shifts by 1, row 2 by 2, and row 3 by 3

70. The purpose of the eight S-boxes in the DES round function is chiefly to:

- A. Perform the initial and final permutations of the 64-bit block
- B. Introduce nonlinearity into the encryption process**
- C. Increase the effective key length beyond 56 bits
- D. Generate the sixteen round subkeys from the original key

Answer: B. Introduce nonlinearity into the encryption process

71. The attack in which an adversary establishes separate shared keys with each party in an unauthenticated Diffie-Hellman exchange, secretly relaying and altering messages between them, is called a ____ attack.

Answer: man-in-the-middle

72. Choosing a very small public exponent such as $e = 3$ without message padding creates a risk known as:

- A. A doubling of the ciphertext length compared to using a larger exponent
- B. An immediate leak of the private exponent d to any observer of the ciphertext
- C. A complete loss of the ability to decrypt the message even with the correct key
- D. A cube-root attack, where the same message sent to multiple recipients can be recovered without factoring n**

Answer: D. A cube-root attack, where the same message sent to multiple recipients can be recovered without factoring n

73. The Chinese Remainder Theorem allows a number to be uniquely reconstructed within a given range from:

- A. Its binary representation alone, without reference to any modulus
- B. A single large modulus formed by adding all the smaller moduli together
- C. Its remainders with respect to a set of moduli that share no common factors with one another**
- D. Its prime factorization expressed in base two

Answer: C. Its remainders with respect to a set of moduli that share no common factors with one another

74. According to NIST SP 800-57 key-size comparisons, achieving a 128-bit symmetric-equivalent security level requires approximately:

- A. A 1024-bit RSA modulus versus a 1024-bit ECC key, since both scale identically
- B. A 512-bit RSA modulus versus a 3072-bit ECC key, the reverse of the usual comparison
- C. The same 128-bit key length for both RSA and ECC, with no difference between them
- D. A 3072-bit RSA modulus versus roughly a 256 to 383-bit ECC key**

Answer: D. A 3072-bit RSA modulus versus roughly a 256 to 383-bit ECC key

75. Given $84 = 2^2 \times 3 \times 7$ and $126 = 2 \times 3^2 \times 7$, $\gcd(84, 126)$ equals:

- A. 21
- B. 42**
- C. 84
- D. 126

Answer: B. 42

76. DES uses an initial permutation, ____ rounds of processing, and then the inverse of the initial permutation to produce the ciphertext.

Answer: 16

77. The block cipher mode that turns a block cipher into a stream cipher by encrypting a counter value for each block and XORing it with the plaintext is called ____ mode.

Answer: CTR

78. The set of points on an elliptic curve, together with a defined addition rule and a point at infinity, forms an algebraic structure called an ____ group.

Answer: abelian

79. Testing whether 29 is prime by checking divisibility only up to $\sqrt{29} \approx 5.4$ works because:

A. Any factor larger than the square root of a number must be paired with a factor smaller than the square root

B. Numbers greater than the square root are automatically assumed to be prime

C. The square root always equals exactly half of the number being tested

D. 29 is a special case, and this shortcut does not generalize to any other number

Answer: A. Any factor larger than the square root of a number must be paired with a factor smaller than the square root

80. The process that stretches the original AES cipher key into a separate 128-bit round key for every round is called key ____.

Answer: expansion

81. The statement $a \equiv b \pmod{n}$ means that:

A. a multiplied by b is always evenly divisible by n

B. n divides (a - b), so a and b leave the same remainder when divided by n

C. a and b are both prime numbers that happen to share the modulus n

D. a and b must be equal integers before applying the modulus operation

Answer: B. n divides (a - b), so a and b leave the same remainder when divided by n

82. The formal statement $a^{(p-1)} \equiv 1 \pmod{p}$, true whenever p is prime and a is not divisible by p, is known as ____ Little Theorem.

Answer: Fermat's

83. Which mode of operation is generally considered to have no safe use case for general-purpose encryption of typical application data?

- A. CTR
- B. ECB**
- C. CBC
- D. OFB

Answer: B. ECB

84. Repeating the Miller-Rabin test with 10 independent random witnesses is done in order to:

- A. Double the effective bit length of the number being tested
- B. Guarantee total mathematical certainty equivalent to a deterministic primality proof
- C. Ensure the candidate number is also proven to be a primitive root
- D. Drive the probability of a false "probably prime" result down to an extremely small chance, suitable for cryptographic use**

Answer: D. Drive the probability of a false "probably prime" result down to an extremely small chance, suitable for cryptographic use

85. The frequency (monobit) statistical test for randomness evaluates a sequence by:

- A. Comparing the sequence against a database of previously observed random sequences
- B. Checking whether the sequence contains any repeated substrings of length four or more
- C. Counting the number of 0s and 1s and checking whether they occur in roughly equal proportion**
- D. Measuring how quickly the sequence can be compressed by a general-purpose algorithm

Answer: C. Counting the number of 0s and 1s and checking whether they occur in roughly equal proportion

86. A function that is easy to compute in one direction but infeasible to invert without secret extra information is called a ____ one-way function.

Answer: trapdoor

87. A sequence produced by a linear congruential generator can pass every standard statistical randomness test and still be considered cryptographically worthless because:

- A. LCG output is always shorter than the plaintext it is meant to protect

B. An attacker who recovers a handful of consecutive outputs can solve for the generator's parameters and predict all future values

C. Statistical tests are incapable of evaluating any algorithmically generated sequence

D. LCGs cannot produce output that is uniformly distributed under any parameter choice

Answer: B. An attacker who recovers a handful of consecutive outputs can solve for the generator's parameters and predict all future values

88. Using the Chinese Remainder Theorem to speed up RSA's private-key operations works by:

A. Splitting the decryption exponentiation into two smaller computations modulo p and modulo q, then recombining the results

B. Replacing modular exponentiation entirely with simple XOR operations

C. Avoiding the need to know either p or q once the public key has been published

D. Reducing the RSA key size requirement from 2048 bits down to 512 bits

Answer: A. Splitting the decryption exponentiation into two smaller computations modulo p and modulo q, then recombining the results

89. In the Micali-Schnorr RSA-based PRNG, each stage encrypts the current internal state and then splits the result so that:

A. Some bits feed back to seed the next stage while the remaining bits form part of the pseudorandom output

B. All of the output bits are discarded and only the feedback bits are ever kept

C. Half of the RSA modulus n is regenerated fresh at every single stage

D. The entire output becomes the feedback value for the next stage, with nothing output

Answer: A. Some bits feed back to seed the next stage while the remaining bits form part of the pseudorandom output

90. The AES round transformation that mixes the four bytes within each column using matrix multiplication over $GF(2^8)$ is called ____.

Answer: MixColumns

91. Using the Euclidean Algorithm on 252 and 105 ($252 = 2 \times 105 + 42$, $105 = 2 \times 42 + 21$, $42 = 2 \times 21 + 0$), $\gcd(252, 105)$ equals:

A. 21

B. 105

C. 7

D. 42

Answer: A. 21

92. A block cipher differs from a stream cipher primarily in that it:

- A. Always runs faster than a stream cipher on the same hardware
- B. Encrypts a fixed-size group of plaintext bits at once, rather than one bit or byte at a time**
- C. Uses a longer key than any stream cipher can support
- D. Cannot be combined with a chaining or feedback mechanism

Answer: B. Encrypts a fixed-size group of plaintext bits at once, rather than one bit or byte at a time

93. The AES-based mode approved by NIST in 2010 specifically for encrypting block-oriented storage devices such as disks is called ____.

Answer: XTS-AES

94. XTS-AES is a specialized mode designed primarily for:

- A. Block-oriented storage devices such as hard drives and SSDs**
- B. Producing message authentication codes for email
- C. Generating session keys for TLS handshakes
- D. Streaming voice and video data over unreliable wireless links

Answer: A. Block-oriented storage devices such as hard drives and SSDs

95. Optimal Asymmetric Encryption Padding (OAEP) is recommended for RSA mainly because it:

- A. Removes the need to generate two distinct large prime numbers
- B. Randomizes the plaintext before encryption, defeating the malleability exploited by chosen-ciphertext attacks**
- C. Allows RSA to be used without ever computing a private exponent d
- D. Increases the size of the RSA modulus n automatically at encryption time

Answer: B. Randomizes the plaintext before encryption, defeating the malleability exploited by chosen-ciphertext attacks

96. The Diffie-Hellman key exchange allows two parties to establish a shared secret key by:

- A. Directly transmitting the shared secret key encrypted under RSA
- B. Meeting in person beforehand to physically exchange the key on paper
- C. Using a trusted third party to generate and distribute the key to both sides

D. Exchanging public values computed from private exponents over an insecure channel, without ever transmitting the secret key itself

Answer: D. Exchanging public values computed from private exponents over an insecure channel, without ever transmitting the secret key itself

97. In a stream cipher, the pseudorandom sequence of bits combined with the plaintext via XOR to produce ciphertext is called the ____.

Answer: keystream

98. RSA's underlying trap-door one-way function relies on the difficulty of:

- A. Finding a collision in a cryptographic hash function
- B. Solving the discrete logarithm problem over a finite field
- C. Reversing a bitwise XOR operation without knowing one of the two inputs

D. Factoring the product of two large prime numbers

Answer: D. Factoring the product of two large prime numbers

99. In a Feistel cipher, the relationship that generates the right half of round i is:

A. $R_i = F(L_{i-1}, K_i) \text{ XOR } R_{i-1} \text{ XOR } L_{i-1}$

B. $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$

C. $R_i = L_{i-1} \text{ AND } F(R_{i-1}, K_i)$

D. $R_i = F(R_{i-1}, K_{i-1}) \text{ XOR } L_i$

Answer: B. $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$
